Weakly Supervised Remote Sensing Image Semantic Segmentation With Pseudo-Label Noise Suppression

Xiao Lu, Zhiguo Jiang, *Member, IEEE*, and Haopeng Zhang, *Member, IEEE*

***Abstract*— Semantic segmentation of remote sensing images (RSIs) plays a crucial role in various applications, including urban planning and environmental monitoring. However, the high cost and complexity of obtaining detailed annotations for RSIs pose significant challenge. This issue necessitates the exploration of weakly supervised learning as an effective alternative, which utilizes more readily available, less granular forms of labeling. Yet, weakly supervised approaches face their own set of challenges, primarily due to scarcity of precise pixel-level labels which significantly hampers the model's ability to learn accurate representations. In this article, we introduce a weakly supervised semantic segmentation (WSSS) approach for RSIs that leverages self-supervised learning (SSL) and pseudo-label noise mitigation to address these challenges. Our method leverages a self-supervised encoder for providing similarity information, which enhances feature representation in RSIs and enables the generation of more accurate pseudo-labels, thus reducing the noise in the pseudo-labels. Furthermore, we propose a refined loss function that incorporates gradient clipping and label smoothing to mitigate the impact of noisy labels, thereby improving the robustness and accuracy of the segmentation results. Extensive experiments on the ISPRS Potsdam, ISPRS Vaihingen, and iSAID datasets demonstrate that our approach achieves state-of-the-art (SOTA) performance, closely matching that of fully supervised methods. Our method not only reduces the dependency on expensive pixel-level annotations but also showcases the potential of SSL in enhancing WSSS tasks.**

***Index Terms*— Image-level label, pseudo-label noise suppression, remote sensing images (RSIs), self-supervised learning (SSL), weakly supervised semantic segmentation (WSSS).**

I. Introduction

REMOTE sensing images (RSIs) contain rich surface feature information. By applying semantic segmentation techniques to RSIs, key information can be extracted automatically, providing practical guidance for people to analyze and utilize land resources rationally [1], [2]. This technology is widely used in various fields such as military [3], agriculture [4], urban planning [5], natural disaster monitoring [6], road analysis [7], and forest vegetation management [8].

RSIs are packed with various information, but the complex structure of cities can easily lead to target occlusion [9], shadows, and other noise. Moreover, RSIs are more prone to object deformation than natural images [10], increasing the difficulty of obtaining accurate information. Recently, relying on a large amount of annotated data, fully supervised semantic segmentation (FSSS) methods such as dc-swin [11], UNetformer [12], and LSKNet [13] have demonstrated superior performance in the field of computer vision. These methods, due to their high generalization and rapid inference capabilities, can effectively extract robust feature information and have been gradually applied to the semantic segmentation tasks of RSIs [14].

However, in the field of remote sensing, although data is easy to obtain, the annotation of labels faces challenges in terms of high difficulty and cost [15]. Therefore, in order to reduce the reliance on pixel-level labels, researchers have begun to explore the use of sparse labeling techniques [16]. This involves learning semantic segmentation methods through points [17], bounding boxes [18], scribbles [19], and image-level labels [20]. In particular, since image-level labels only need to indicate the category information of the target object, annotation cost is the lowest, and weakly supervised semantic segmentation (WSSS) methods based on image-level labels have received widespread attention.

Currently, although a variety of WSSS models have been developed in the field of natural images, these models perform poorly when directly applied to remote sensing data [21]. As shown in Fig. 1, this is mainly due to the inherent traits of remote sensing data, which includes challenges in locating small targets, significant variability within classes, and minor distinctions between categories [22]. These factors result in a lack of sufficient supervisory information for these models when processing remote sensing data, thereby affecting the accuracy of pseudo-labels. Consequently, this has led to the majority of past studies being focused solely on tasks within a single category [23], [24], failing to satisfy the practical demands. In response, self-supervised learning (SSL) methods such as MAE [25], DINO [26], and iBOT [27], by using the input data itself as supervisory information without the need for manual annotation, can effectively improve the quality

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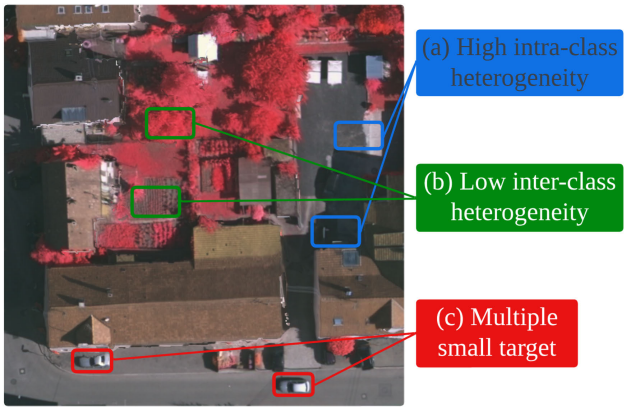


Fig. 1. Illustration of challenges in weakly supervised remote sensing image semantic segmentation. (a) Different regions of the impervious surfaces class showing variability in texture and appearance. (b) Instances of the low vegetation class with minor distinctions from tree instances. (c) Multiple small target cars in the scene make it difficult to obtain accurate location information.

of encoder features, showing a natural advantage in handling unlabeled tasks [28]. Based on this, our study first constructs a classification model optimized by feature similarity obtained from self-supervised encoder to generate a more accurate pseudo-label, thereby suppressing the noise in the pseudo-label. Then, by applying gradient clipping to optimize the segmentation loss function and adopting label smoothing to reduce the impact of noisy labels, we can gain a more robust and accurate segmentation result after iterative training. The main contributions can be summarized as follows.

1) To address the complexity of information and the lack of supervised knowledge in RSIs, we designed a module that optimizes the computation of feature relationships using similarities obtained from a fine-tuned self-supervised encoder, thereby supplementing semantic information and resulting in more accurate pseudo-labels.
2) Due to the insufficient constraints of RSIs, which lead to incorrect cases in pseudo-labels, we propose a loss function using gradient clipping and label smoothing, aiming to reduce the adverse impact of noisy labels on the training phase of the segmentation network.
3) We propose a WSSS method for RSIs, achieving state-of-the-art (SOTA) results on the ISPRS Potsdam, ISPRS Vaihingen, and iSAID datasets.

Subsequently, in Section II, we provide a concise overview of the related work in the domain of WSSS for RSI and SSL. In Section III, we provide a detailed description of the WSSS method we propose. Section IV presents extensive experiments conducted on four datasets, demonstrating the superiority of our approach. Conclusions are articulated in Section V.

II. Related Works

A. WSSS for RSI

Due to the high cost of full supervision labeling, WSSS is getting more attention [29]. Weakly supervised methods, especially those based on image-level labels, suffer from a lack of pixel-level labels so that the supervision information is limited to the existence of target categories without any location or contour information, which significantly differs from fully supervised methods in terms of supervision information. A common solution is to use class activation maps (CAMs) [30] to generate pseudo-pixel-level labels to train the segmentation model, identifying target areas through classifier weight activation. However, CAM often only covers the most prominent areas and may include non-target areas, leading to inaccurate pseudo-labels and affecting segmentation performance [31]. Therefore, optimizing CAM to obtain more accurate pseudo-labels is key in WSSS. Common methods include randomly masking images to encourage the network to focus on various areas [32], using pixel relationships to expand CAM seeds [33], and improving feature quality through equivariance [34]. Due to differences in data characteristics, the focus of research on weakly supervised methods for RSIs diverges. Wang et al. [35] accomplished crop segmentation by applying thresholds to binary CAM. Cheng et al. [36] pioneered the application of WSSS for mapping woody vegetation using image-level labels by a dual-branch network. Qiao et al. [37] present a WSSS model to extract pixel-wise damaged buildings by integrating the special characteristics of damaged buildings into network. Zhou et al. [22] introduced a self-supervised Siamese network, processing data with explicit pixel-level constraints, marking the first method to use image-level labels for multiclass WSSS of RSIs. This method improves semantic feature quality by constructing positive and negative samples, introduces confidence-weighted calculation of the class token to reduce noise interference, and combines feature fusion to optimize the initial CAM, generating refined pseudo-labels for training. CISM [38] explores cross-image semantics in RSIs with two novel loss functions to obtain better results. Considering the unique characteristics of remote sensing data, OME [39] introduces category label optimization strategies. It leverages the category cross correlation matrix to better filter out improper training samples. Combined with the URN method [40], it adjusts the segmentation network loss function through confidence weighting to reduce noise impact and improve the robustness of segmentation results. However, these approaches fail to incorporate adequate additional semantic information when generating pseudo-labels, particularly the segmentation of small targets still requires further improvement.

B. SSL

SSL is an approach to representation learning in which a model learns to capture the relationships between its input data. This method allows the model to leverage large amounts of unlabeled data to understand underlying patterns and structures through pretext tasks. MAE [25] reconstructs missing parts of the input data as a pretext task, which helps the model develop robust feature representations. iBOT [27] employs masked image modeling to dynamically adjust input tokens, enhancing the model's ability to learn meaningful representations. DINO [26] uses vision transformers (ViTs) [41] in contrastive learning that performs self-distillation with exponential moving average updates [42]. Despite the absence

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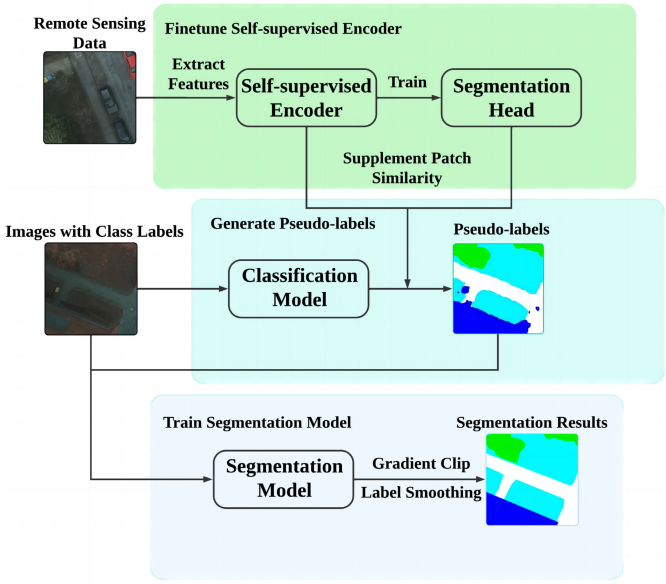


Fig. 2. Overview of our proposed WSSS method.

of any annotated information during the training process, the resulting image features explicitly contain positional information, particularly regarding object boundaries. Furthermore, these image features demonstrate a high degree of semantic consistency, both within and between images [43].

Consequently, SAS [44] utilizes features generated by DINO as guidance for superpixels, directing the growth of seed regions within CAM, thereby obtaining pseudo-labels of higher accuracy. This method combines the similarity and positional information of DINO features for clustering and scores the respective regions based on the response intensity of CAM. Through iterative optimization, the final results are obtained. The guidance of CAM seed region growth by DINO feature similarity enhances accuracy, fully demonstrating the feasibility of using the pretrained model obtained by SSL to provide additional information to enhance pseudo-labels. However, this approach cannot guarantee the accuracy of DINO feature similarity on specific datasets, and therefore the segmentation results may not be satisfactory.

III. Method

This section introduces the complete workflow of our method as shown in Fig. 2, which encompasses fine-tuning self-supervised encoder on remote sensing data, the process of generating CAMs with additional information from fine-tuned self-supervised encoder, and achieving robust segmentation network training using gradient clipping and label smoothing.

A. Fine-Tune Self-Supervised Encoder on Remote Sensing Data

Self-supervised learning has emerged as a powerful paradigm in computer vision, enabling models to learn useful representations from unlabeled data by solving pretext tasks. These pretext tasks, such as distinguishing between different augmentations of the same image [26] or reconstructing masked parts of image [25], allow the encoder to capture rich semantic features without requiring manual annotations. Although the features obtained by self-supervised encoder can explicitly reveal the semantic segmentation information of objects in images, making it suitable for numerous computer vision tasks, it is primarily trained on natural images. This exhibits significant distribution differences from RSIs, which contain richer semantic information, smaller inter-class variance, and fewer background regions, leading to a degradation in model accuracy [45]. Therefore, we trained a segmentation head using unsupervised semantic segmentation that maps natural images features onto the feature space of remote sensing data. The objective of the segmentation head is to learn a nonlinear projection, which reduces the impact caused by the distribution shift from natural images to RSIs.

Given an image processed to self-supervised encoder, we have feature vector $f_i \in \mathbb{R}^C$ corresponding to patch i of input image, the segmentation head maps this feature to $s_i \in \mathbb{R}^k$ and the projector further project patch to $z_i \in \mathbb{R}^k$. The segmentation head is constructed with a two-layer relu MLP, and the projection head is composed of a linear layer equipped with a normalization layer. We only use the projector during the training stage. In the inference stage, we use the segmentation feature s_i. In order to ensure the semantic consistency of the projected features, contrastive loss can be defined as

$$L_i = -\log \frac{\exp\left(\text{sim}\left(z_i, z_j\right)/\tau\right)}{\sum_{a\in A}\exp(\text{sim}(z_i, z_a)/\tau)} \tag{1}$$

where A indicates all indexes except i, j is the index of positive pairs of ith patch, τ is temperature parameter, and $\text{sim}(\cdot,\cdot)$ denotes similarity metrics. In this study, positive pairs refer to samples with semantically similar information, which are selected based on a feature similarity threshold. This article employs cosine similarity as the metric for measuring similarity, which is defined as

$$\text{sim}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|}. \tag{2}$$

The most important part of the fine-tuning process is how to select accurate positive sample pairs in the absence of precise supervision information. HP [46] chooses positive sample pairs by comparing similarities with a candidate pool. By setting a threshold c calculated from the candidate pool, it is ensured that only those samples with a high degree of similarity are considered positive, thereby improving the accuracy and robustness of the model. However, we consider that operations with the candidate pool sometimes lead to inaccurately selected positive samples due to excessively high similarity within the pool. Additionally, in RSIs, there is the issue of excessively high feature similarity between adjacent regions of different categories, which can interfere with the positive pairs selection process. To address these challenges, we propose incorporating the average value within a batch, which can dynamically adjust the selection criteria based on the overall distribution of feature similarities, leading to more effective and adaptive positive sample selection.

Specifically, we can get a reference pool $q_m \in Q^{ag}$ by randomly sampling patch features f_i. Then, we define a

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reference pool threshold c_i to collect positive samples as

$$c_i = \max_{q_m \in Q^{ag}} \text{sim}(q_m, f_i). \tag{3}$$

For each features, we can get its positive samples z_j by reference pool threshold c_i and overall average value $\bar{f}$ as

$$z_j = \left\{ j | \text{sim}\left(f_i, f_j\right) > \min\left(c_i, c_j\right) \cap \text{sim}\left(f_i, f_j\right) > \bar{f} \right\}. \tag{4}$$

The overall average value $\bar{f}$ is defined as

$$\bar{f} = \frac{1}{|P|} \sum_{(i,j) \in P} \text{sim}\left(f_i, f_j\right) \tag{5}$$

where $P = \{(i, j) \mid 1 \leq i < j \leq N\}$ indicates the set of all distinct patch pairs within a batch. After constructing the positive sample pair relationship based on the threshold, the training process of the segmentation head is supervised by the contrastive loss to achieve the self-supervised encoder domain adaptation process from natural data to remote sensing data.

B. Generating Class Activation Map

In the prediction phase of the classification network, CAM can effectively extract the target location information. Due to its effectiveness and simplicity, CAM has been widely used to generate initial pseudo-labels for WSSS.

Specifically, given an image, its feature map $F \in \mathbb{R}^{hw \times d}$ is extracted with classification encoder, where hw and d denote the number of spatial and channel dimensions. The CAM is computed by weighted sum of the feature maps with classification layer $W \in \mathbb{R}^{c \times d}$, where c is the number of classes. Then, relu and normalization are applied to ensure the proper range of CAM. Therefore the CAM for class c is defined as

$$\text{CAM}_c(F, W) = \frac{\text{relu}(M_c)}{\max\left(\text{relu}(M_c)\right)}, \ : M_c = \sum_i W_{c,i} F_{:,i}. \tag{6}$$

CAM suffers from over-smoothing when extracting features with ViT [47], affecting the accuracy of the activation regions. To address this, Toco [48] proposes a patch token contrast (PTC) module to supervise the token relations in the output layer by using the feature information from the intermediate layers, improving the quality of the output layer features. However, this method relies on the intermediate layers for supervisory information, lacks additional semantic prior knowledge, and is prone to positive feedback loops. Inspired by SAS [44], we propose a module that use fine-tuned self-supervised encoder to supplement the missing accurate location information, achieving more robust and precise token relations, thereby reducing the noise in the pseudo-labels. Fig. 3 shows our workflow for optimizing pseudo-label.

To extract semantically aware knowledge from intermediate layer features F^m, we add an auxiliary classification layer W^m. Therefore, we can get intermediate CAM M^m as

$$M^m = \text{CAM}\left(F^m, W^m\right). \tag{7}$$

After obtaining the CAM of the middle layer, it can be converted into pseudo-label Y_i by the threshold β_l to supervise the token relations of the output layer. When constructing the PTC loss, patch tokens with the maximum activation value in the CAM less than the threshold are ignored, and only the token relations between the more reliable target areas are considered. This process can be described as

$$Y_i = \begin{cases} \text{Arg}\max\left(M^{(i,:)}\right), & \text{if } \max\left(M^{(i,:)}\right) > \beta_l \\ 255, & \text{if } \max\left(M^{(i,:)}\right) < \beta_l. \end{cases} \tag{8}$$

Specifically, if two tokens share the same semantic label, they are labeled as positive pairs; otherwise, they are labeled as negative pairs, thereby constructing token relations for the PTC loss. To correct token relationships, patch i is put into self-supervised encoder to obtain corresponding feature S_i. Positive samples are then selected based on similarity. In particular, if the similarity between two patch tokens in self-supervised encoder feature space is greater than the average in the batch, they are regarded as positive examples, as

$$Y_i = Y_j, \ if \ \text{sim}\left(S_i, S_j\right) > \frac{1}{|P|} \sum_{(k,l) \in P} \text{sim}(S_k, S_l) \tag{9}$$

where $P = \{(k, l) \mid 1 \leq k < l \leq hw\}$ indicates the set of all distinct patch pairs within a batch.

In summary, if patch token similarity in the self-supervised encoder's feature space exceeds the batch average, they are considered as positive pairs. If their corresponding similarity does not exceed the average, the semantic labels of the middle layer are further compared. If they are the same, they are also considered positive pairs; if different, they are deemed negative pairs. Based on these positive and negative pairs, PTC loss is then constructed as

$$L_{ptc} = \frac{1}{N^+} \sum_{Y_i = Y_j} \left(1 - \text{sim}\left(F_i, F_j\right)\right) + \frac{1}{N^-} \sum_{Y_i \neq Y_j} \text{sim}\left(F_i, F_j\right). \tag{10}$$

By adopting the feature similarity obtained from self-supervised encoder to construct the relationship of sample pairs, this module not only supplements the constraint knowledge missing from the semantic labels of ignored areas but also alleviates the issue of token relationship misalignment caused by the positive feedback effect of semantic label information during the training process. Considering that the possible negative similarity values might interfere with the loss function and average value metrics [49], we adopt the absolute value of cosine similarity to address this issue.

C. Segmentation With Noisy Labels

Due to the noise resistance of deep learning, we adopt an iterative training method for training the segmentation network, using the results obtained from the segmentation network as pseudo-labels for the next round of iterative training [50]. The pseudo-labels obtained from CAM inevitably contain noise. Directly using these pseudo-labels as annotations to train the segmentation network would introduce amplified noise interference, affecting the final segmentation results.

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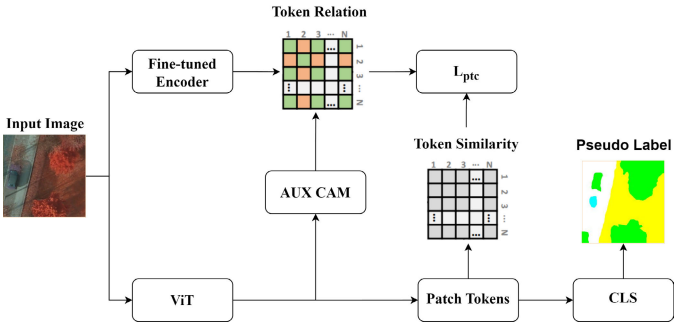


Fig. 3. Illustration of pseudo-label refinement utilizing token relations derived from fine-tuned self-supervised encoder. We obtain token relations using feature similarity from a fine-tuned self-supervised encoder and semantic labels from auxiliary CAM in the intermediate block (AUX CAM). The token relations are combined with output patch tokens to construct the PTC loss, which improves the semantic quality of the patch tokens. Pseudo-labels are generated with patch tokens using a classifier (CLS).

To address this, we propose a new loss function with gradient clipping and label smoothing to reduce the noise's impact on the segmentation result. Previous works have pointed out that noisy labels often come with larger gradients [51], [52]. Therefore, we use a method that considers both global average and local gradients to select an appropriate threshold for gradient clipping. Specifically, we divide the segmentation results into L^2 non-overlapping patches, calculate the cross-entropy (CE) loss of segmentation results P_i with CAM_i, and select the maximum value between the CE loss of the patch and the global average as the threshold θ_i. This process can be described as

$$\lambda_i = CE(P_i, C\ AM_i), i \in \left\{1, \cdots L^2\right\}$$
$$\theta_i = \max\left((\lambda_i), \frac{1}{L^2}\sum_{i=1}^{I^2}\lambda_i\right). \tag{11}$$

Thus, by selecting the maximum value between the loss of the patch and the average loss of the entire image as the threshold, a truncation operation is performed on all losses within the patch which can be described as

$$L_{\text{seg}} = \begin{cases} \lambda_i, & \text{if } \lambda_i \le \theta_i \\ \theta_i, & \text{if } \lambda_i > \theta_i. \end{cases} \tag{12}$$

This reduces the impact of noisy labels and their associated excessive loss functions on the segmentation network. Since the segmentation network tends to learn from clean labels in the early stages of training, quickly reducing loss while significantly improving segmentation accuracy, we choose to perform gradient clipping only when the average loss is below a hyperparameter, ensuring that early learning is unaffected.

On the other hand, research has shown that label smoothing can reduce the confidence of noisy labels while allowing the noise in label smoothing to compete with the noise in annotations [53], thereby reducing the harm of incorrect labels to the training process. Therefore, our segmentation loss adopts CE loss with label-smoothing parameter $\epsilon = 0.1$ as the loss function which means we replace the original label distribution $q(k) = \delta_{k,y}$ with

$$q'(k) = (1-\epsilon)\delta_{k,y} + \frac{\epsilon}{K}. \tag{13}$$

Thus, the CE loss with label smoothing can be written as

$$CE = -\sum_{k=1}^{K} q'(k)\log(p(k)) \tag{14}$$

where $\delta_{k,y}$ is the Kronecker delta, which is 1 if $k = y$ and 0 otherwise, K is the total number of classes, and $p(k)$ is the segmentation result.

IV. Experiments

To verify the effectiveness of our method, we conducted extensive experiments on remote sensing semantic segmentation datasets ISPRS Potsdam and Vaihingen [56], as well as the iSAID datasets [57]. To fully demonstrate the superiority of our approach, we also experimented on the natural semantic segmentation dataset PASCAL VOC 2012 [58] and compared it with other methods. The results confirmed that our method could achieve SOTA.

A. Datasets and Evaluation Metrics

1) ISPRS Potsdam Dataset: This dataset was collected in the city of Potsdam and consists of 38 orthoimages with a spatial resolution of 5 cm, each measuring 6000 × 6000. Following the same processing approach as OME [39], we ignore the clutter category, so the dataset comprises five categories: impervious surfaces, buildings, low vegetation, trees, and cars. Following the official data division method, 23 images are divided into the training sets and 14 as the validation sets, with images 7–10 excluded due to incorrect labeling. Unlike OME [39], we directly crop the images into non-overlapping 256 × 256 pictures because the presence of additional information about target categories during the data filtering process does not align with the original intention of WSSS, resulting in 13 248 training images and 8064 validation images.

2) ISPRS Vaihingen Dataset: This dataset was collected in Vaihingen, a small town, and consists of 33 orthophotos of 2494 × 2064 with a spatial resolution of 9 cm. The category composition is the same as Potsdam. Following the official data division method, 16 pictures are divided into training sets and 17 pictures are divided into validation sets. Similar to Potsdam, we directly crop the images into non-overlapping 128 × 128 images, finally obtaining 4981 training images and 5684 verification images.

3) iSAID Dataset: This dataset was collected from Google Earth, which were mainly taken by the JL-1 and GF-2 satellites. This dataset comprises 15 target categories and one background (BG) category. The target categories specifically include plane (PL), baseball diamond (BD), ship (SH), ground-track field (GTF), basketball court (BC), soccer ball field (SBF), roundabout (RA), storage tank (ST), bridge (BR), large vehicle (LV), tennis court (TC), helicopter (HC), harbor (HA), small vehicle (SV), and swimming pool (SP). According to the original division of the dataset, there are 1411 high-resolution images for training; the remaining 458 images are used as

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TABLE I
COMPARISON OF PSEUDO-LABELS WITH OTHER METHODS ON THE ISPRS POTSDAM DATASET

Method	Imp.Surf.	Building	Low Veg.	Tree	Car	mIoU	Training Time
SEAM [34]	66.65	76.80	63.61	64.63	36.62	61.66	112 mins
Puzzle-CAM [54]	72.19	84.67	65.32	64.82	43.80	66.16	207 mins
AFA [55]	75.15	83.86	67.46	65.42	68.03	71.98	172 mins
Toco [48]	78.98	**90.82**	67.64	66.41	61.26	73.02	83 mins
Ours	**82.87**	88.23	**71.03**	**67.49**	**83.49**	**78.62**	**81 mins**

TABLE II
COMPARISON OF PSEUDO-LABELS WITH OTHER METHODS ON THE ISPRS VAIHINGEN DATASET

Method	Imp.Surf.	Building	Low Veg.	Tree	Car	mIoU
SEAM [34]	70.60	75.78	56.12	67.86	36.79	61.43
Puzzle-CAM [54]	72.49	79.53	52.42	65.18	38.36	61.60
AFA [55]	70.45	79.74	60.59	73.60	27.99	62.47
Toco [48]	**77.90**	79.90	53.05	65.71	34.05	62.12
Ours	76.14	**82.92**	**62.18**	**72.55**	**52.75**	**69.31**

the validation sets. Similar to Potsdam, we directly crop the images into non-overlapping 256 × 256 images, finally obtaining 156 019 training images and 53 176 verification images.

4) PASCAL VOC 2012: This dataset is a classic natural image dataset for object detection and semantic segmentation, mainly comprising 20 foreground categories and one background category. Following the general approach, we utilize supplementary data from SBD, thereby obtaining 10 582 images for training, 1449 images for validation, and 1456 images for testing.

5) Evaluation Metrics: Based on the above datasets, we mainly use the mean intersection over union (mIoU) as the evaluation metric to assess the experimental results. According to the convention, we also use true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs) to express mIoU. Specifically, the mIoU is defined as

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{n}\frac{\text{TP}}{\text{TP+FP+FN}}. \tag{15}$$

B. Implementation Details

For self-supervised encoder, we employ DINO pretrained ViT-base models. The feature dimension output by the segmentation head is 512. For the part of training the classification network to generate CAM, we chose ViT-base as the backbone network. The patch size is 16, the optimizer is AdamW, the learning rate is 6e-5, the decay rate is 1e-2, and the beta parameters are 0.9 and 0.999. The warm-up learning rate is 1e-6, with training iterations on Potsdam, Vaihingen, iSAID, and VOC12 being 8000, 4000, 40 000, and 20 000, respectively, and the corresponding warm-up training rounds are 1500, 500, 2000, and 1500. The ratios for generating CAM at multiple scales are (1.0, 0.5, 1.5). For the part about training the segmentation network using pseudo-labels, we use deeplabv3+ [59] for the remote sensing datasets. For natural images, to achieve better segmentation accuracy, we use Segmenter [60], with a training duration of 20 k rounds, a batch size of 20, using the SGD optimizer with an initial learning rate of 0.005, input image size of 256, gradient clipping starting when the average loss is less than 0.5, and the segmentation patch size is 32. All experiments were conducted on 2 NVIDIA RTX 4090 GPUs running on Ubuntu 18.04, using PyTorch 1.9 and Python 3.8.

C. Performance

In this part of the experiment, we will compare our method with other remote sensing WSSS methods as well as SOTA WSSS methods for natural images, in terms of evaluation metrics and visual results, to assess the quality of CAMs generated by different methods. At the same time, we also evaluate the differences between the segmentation results obtained using pseudo-labels and those obtained from fully supervised training of the segmentation network, to verify the effectiveness of our method.

1) CAM Performance:

a) ISPRS dataset: The results of the experiments are shown in Tables I and II. The final segmentation results on remote sensing data are shown in Tables III and IV. In remote sensing data, we compare our proposed method with SEAM mentioned in OME, puzzle-cam, and the WSSS methods for natural images, AFA, and Toco. SEAM only uses equivariant constraints and pixel relationships to provide additional supervisory information to optimize the CAM obtained by the convolutional network, lacking sufficient additional semantic supervisory information and contextual knowledge, resulting in the worst performance. AFA and Toco methods, on the other hand, extract contextual information through transformers and combine it with the attention matrix, significantly improving the accuracy of the pseudo-labels obtained through improvements to the backbone network and the additional constraint information provided by the attention matrix. On this basis, the

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TABLE III
COMPARISONS OF SEGMENTATION RESULT ON THE ISPRS POTSDAM AND VAIHINGEN DATASETS WITH OTHER METHODS

Dataset	Strategy	Method	Imp. Surface	Building	Low Veg.	Tree	Car	mIoU
Potsdam	FSSS	Deeplabv3+	89.45	94.44	77.87	77.98	94.02	86.75
	WSSS	Ours	84.70	89.81	73.70	73.02	86.92	81.63
	WSSS	OME [39]	74.17	82.40	69.77	71.96	45.94	68.85
Vaihingen	FSSS	Deeplabv3+	85.52	90.91	69.78	79.39	78.11	80.74
	WSSS	Ours	78.04	83.44	58.09	70.24	62.71	70.50
	WSSS	OME [39]	75.15	79.11	56.00	69.99	54.35	66.92
	WSSS	S^2EPC [22]	53.86	53.43	44.40	55.79	19.66	45.43

TABLE IV
COMPARISONS OF SEGMENTATION RESULT ON THE ISAID DATASET WITH OTHER WSSS METHODS

Method	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU
OME [39]	75.83	15.23	46.13	4.21	6.13	7.48	16.23	21.31	12.36	9.24	53.18	26.29	41.27	6.13	2.34	5.18	23.26
CISM [38]	83.11	35.51	44.63	11.87	19.51	8.42	48.36	34.42	29.21	30.55	52.63	31.91	44.18	10.82	17.63	14.34	32.23
S^2EPC [22]	86.11	**38.30**	**46.70**	11.48	17.10	8.71	**53.30**	36.68	**34.59**	31.70	56.74	30.20	49.10	11.59	19.00	15.86	36.79
Ours	**98.02**	15.23	17.26	**34.94**	**21.06**	**20.63**	32.36	**47.01**	13.39	**67.69**	**63.17**	**49.21**	**53.97**	**37.66**	**31.95**	**34.29**	**39.86**

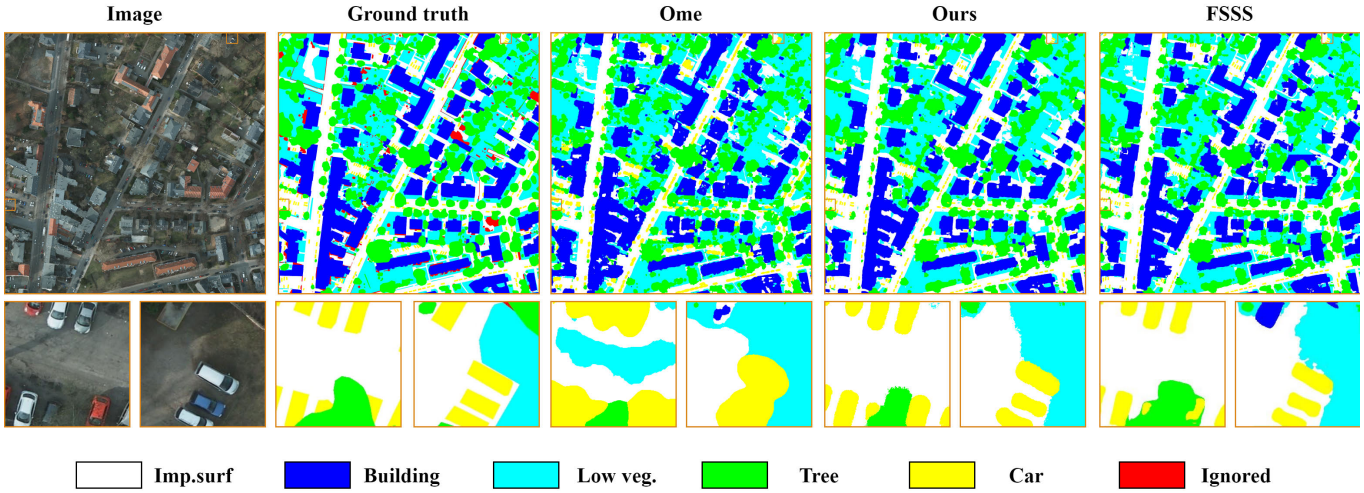


Fig. 4. Examples of segmentation results from WSSS and FSSS methods on the ISPRS Potsdam dataset.

results show that our proposed method is clearly superior to previously proposed WSSS methods for RSIs, outperforming past algorithms in all classification categories on the ISPRS dataset and generating more accurate pseudo-labels. Notably, for difficult small targets like vehicles in complex remote sensing scenes, our method has shown the most significant improvement, with classification accuracy on the Potsdam dataset even surpassing some other categories. This fully reflects that our proposed idea of supplementing similarity information from self-supervised encoder provides additional semantic prior information, compensating for the lack of positional supervisory knowledge in WSSS of RSI, and meeting the semantic segmentation requirements of complex scenes in remote sensing tasks. On the other hand, our method does not use pixel-level optimization during the training process. Instead, it enhances feature relationships by incorporating additional information and improves feature quality through the PTC loss. As a result, the training time is shorter compared to other methods.

b) PASCAL VOC 2012: To prove the generalization of our method and facilitate a fair comparison with other competitive image-level WSSS approaches, we conducted further experiments on the natural image dataset VOC12. As shown in Table V, our method outperforms others, thereby showcasing its strong generalizability.

2) Segmentation:

a) ISPRS dataset: In our study, we employed pseudo-labels generated in the first stage as initial labels to supervise the training process of the segmentation network. We compared the segmentation results with FSSS methods and WSSS methods for remote sensing data. As shown in Table III, the results clearly demonstrate that our method significantly

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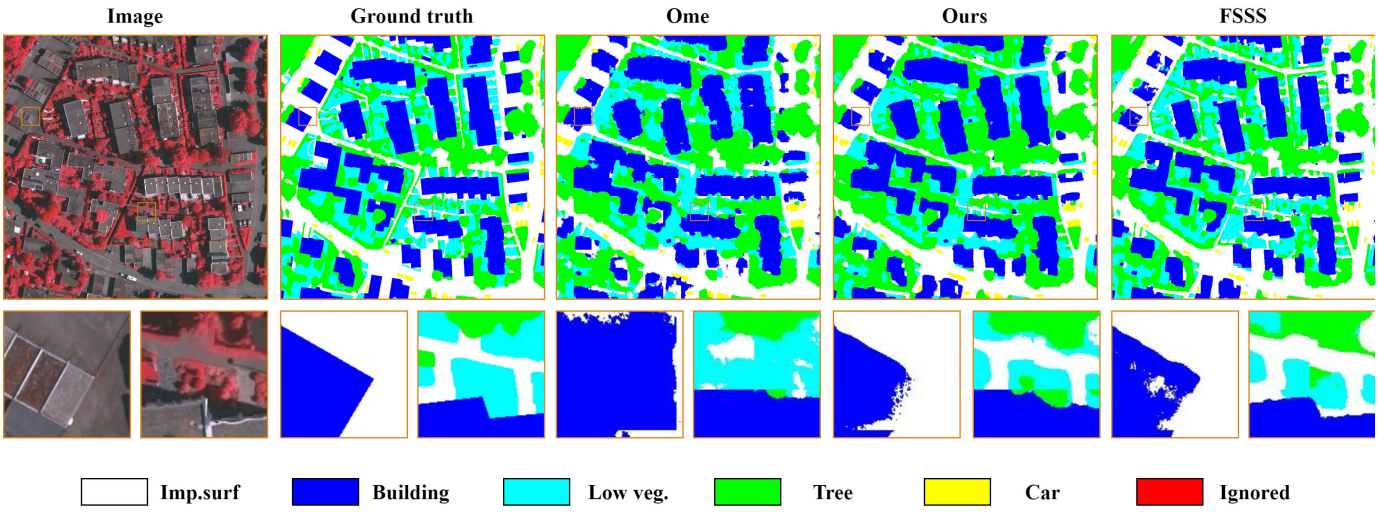


Fig. 5. Examples of segmentation results from WSSS and FSSS methods on the ISPRS Vaihingen dataset.

TABLE V
COMPARISON OF PSEUDO-LABELS WITH OTHER SOTA METHODS ON THE PASCAL VOC 2012 DATASET IN MIOU (%)

Method	Backbone	Val	Test
SEAM [34]	ResNet38	64.5	65.7
Puzzle-CAM [54]	ResNet101	66.9	67.7
AFA [55]	MiT-B1	66.5	68.7
SLRNet [61]	ResNet38	66.2	67.1
ViT-PCM [62]	ViT-B	69.3	71.4
Toco [48]	ViT-B	72.3	73.6
WeakTr [63]	ViT-S	66.2	68.4
Ours	**ViT-B**	**74.3**	**75.7**

TABLE VI
COMPARISON ANALYSIS RESULTS WITH OTHER SOTA METHODS ON THE PASCAL VOC 2012 DATASET IN MIOU (%). SUP. DENOTES THE SUPERVISION TYPE. I: IMAGE-LEVEL LABELS; S: SALIENCY MAPS; B: BOUNDING BOXES; F: FULLY SUPERVISED

Method	Backbone	Sup.	Val	Test
Deeplabv3plus [59]	ResNet50	F	77.5	78.4
Segmenter [60]	ViT-S	F	83.1	82.6
BCM [64]	ResNet101	I+B	70.2	
BBAM [65]	ResNet101	I+B	73.7	73.7
EPS [66]	ResNet101	I+S	71.0	71.8
L2G [67]	ResNet101	I+S	72.1	71.7
AdvCAM [68]	ResNet101	I	67.5	67.1
S^2EPC [22]	ResNet101	I	67.6	67.8
ViT-PCM [62]	ResNet101	I	70.3	70.9
CLIP-ES [69]	ResNet101	I	73.8	73.9
MCTformer [70]	ResNet38	I	71.9	71.6
1Stage [71]	ResNet38	I	62.7	64.3
Toco [48]	ViT-B	I	69.8	70.5
WeakTr [63]	ViT-S	I	78.4	79.0
Ours	ViT-B	I	**79.1**	**79.8**

outperforms other WSSS approaches, with improvements of 12.78% and 3.58% in mIoU on two datasets, respectively. Meanwhile, for the segmentation of small targets, such as cars, the results improved by 40.98% and 8.36%, respectively. This further demonstrates that the additional knowledge provided from self-supervised encoder can ensure the quality of features. Moreover, we attained accuracies of 94.1% and 87.3% relative to fully supervised training, proving the feasibility of using only classification labels for WSSS in remote sensing data tasks, where annotated data are difficult to obtain. To visually compare the segmentation results of different methods, we show the segmentation results on the Potsdam and Vaihingen datasets in Figs. 4 and 5. It can be clearly seen that OME is difficult to ensure the completeness of the segmented objects and the accuracy of edge localization due to its lack of sufficient semantic knowledge. Correspondingly, our method, by incorporating additional semantic information, significantly enhances the precision of boundary segmentation.

b) iSAID dataset: As shown in Table IV, our method achieves better results compared to other WSSS approaches for RSIs on the iSAID dataset. The segmentation result has improved across most categories, with particularly significant enhancements observed in the SV and PL categories, which have increased by 23.46% and 35.99%, respectively. This fully demonstrates the effectiveness of our method in enhancing location information, showcasing its capability to achieve SOTA results on remote sensing data.

c) PASCAL VOC 2012: As shown in Table VI, our model outperforms other WSSS approaches on the VOC12 dataset, fully demonstrating the advantages and generalizability of our method. Remarkably, our approach even surpasses fully supervised methods. The results reveal that our method, without requiring any pre-provided additional annotation information, can obtain sufficient extra supervisory information from self-supervised encoder, ensuring that the segmentation results achieve SOTA performance.

D. Ablation Studies

In order to further verify the effectiveness of our proposed method and select reasonable hyperparameters, we conducted

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TABLE VII
ABLATION STUDIES FOR EACH PART OF CAM GENERATION

Baseline	PTC	SSL	Fine-tune	CAM	AUX CAM
✓				61.30	65.61
✓	✓			73.30	69.95
✓	✓	✓		77.40	68.27
✓	✓	✓	✓	**78.62**	**71.51**

TABLE VIII
ABLATION STUDIES FOR EACH PART OF SEGMENTATION

Baseline	Label Smooth	Gradient Clip	Iterative Train	SEG mIoU
✓				79.85
✓	✓			80.28
✓	✓	✓		80.51
✓	✓	✓	✓	**81.63**

corresponding ablation experiments to verify the effect. We chose to conduct experiments on the Potsdam dataset to demonstrate the module effect on remote sensing data.

1) Generating Pseudo-Labels: In the process of generating pseudo-labels, we adopted ViT-B as the baseline. To better illustrate the effectiveness of our method, we compared the quality of CAM between the output block and the ninth block (AUX CAM). As demonstrated in Table VII, without other constraints, CAM generated by the output layer is overly smooth, leading to a lower quality of CAM at the output layer compared to that from the intermediate layer. After introducing the PTC module, the mIoU of CAM increased by 12%, and surpassed that of AUX CAM, indicating that leveraging high-confidence features from intermediate layer can mitigate the smoothing issue of deep layer features, thereby enhancing the accuracy of pseudo-labels. With the incorporation of SSL information, the quality of CAM further improved by 0.9%, proving that even without fine-tuning on remote sensing data, a self-supervised encoder pretrained on large-scale datasets can provide semantic prior knowledge helpful for precise feature matching. After fine-tuning the self-supervised encoder on remote sensing data, the mIoU increased by an additional 1.2%, overall exceeding the baseline by 17.3%. This demonstrates our method's ability to provide additional supervisory information for obtaining more accurate pseudo-labels.

As shown in Fig. 6, due to the complexity of the scenes, the directly generated CAM failed to accurately capture the locations of vehicles, displaying chaotic activation maps. The situation improved after incorporating the PTC module, but the accuracy was still insufficient. After the addition of SSL similarity knowledge, the activation maps accurately depicted the locations of cars, showcasing the effectiveness and advantages of the module.

2) Segmentation: The experimental results of segmentation ablation studies are shown in Table VIII. We selected CE loss as the baseline. By incorporating label smoothing, the segmentation results improved by 0.43%, demonstrating its ability to reduce the interference of noise in pseudo-labels during training. After introducing gradient clipping, by filtering out potentially noisy gradients, the segmentation mIoU was further enhanced by 0.23%. Due to the noise resistance of the neural network, we use the segmentation results as pseudo-labels for three rounds of iterative training, achieving the final experimental results, which showed an improvement of 1.78% compared to the segmentation network trained directly with CE loss.

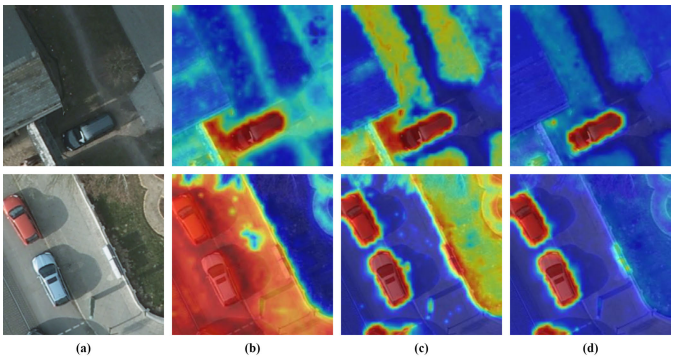


Fig. 6. Visualization results of car's CAM on the ISPRS Potsdam dataset after different modules are added to the baseline. (a) Input image. (b) Baseline. (c) Adding PTC loss. (d) Leveraging similarity information from self-supervised encoder.

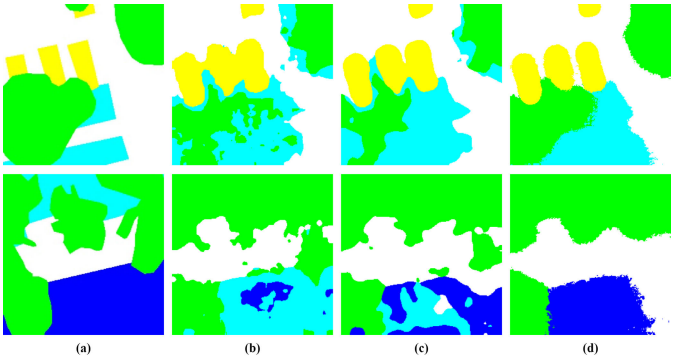


Fig. 7. Visualization results of optimized segmentation. (a) Ground truth. (b) Pseudo-label. (c) WSSS result. (d) FSSS result.

As shown in Fig. 7, the initial pseudo-labels, due to severely insufficient supervisory information, were disturbed by noise from incorrect labels. Directly using them as labels for training with CE loss still retained noise, showing weak noise resistance. However, the results learned from the loss function with added gradient clipping and label smoothing, although still containing errors, were able to effectively reduce the impact of incorrect labels on the gradients, thereby enhancing robustness and improving segmentation results.

3) Self-Supervised Encoder: To investigate the encoding efficacy of various self-supervised methods, this study compares the CAM results of patch token relations optimized by different self-supervised encoders, aiming to identify the optimal self-supervised encoder for this specific scenario. The results are shown in Table IX.

By incorporating additional information through self-supervised encoders, the quality of CAM is enhanced across

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TABLE IX
COMPARISON OF CAM AND AUX CAM MIOU OPTIMIZED USING DIFFERENT SELF-SUPERVISED ENCODERS

Method	Backbone	CAM	AUX CAM
Baseline	-	73.30	69.95
iBOT [27]	ViT-B	74.36	70.90
MAE [25]	ViT-B	75.24	68.11
DINO [26]	ViT-S	72.39	67.33
DINO [26]	ViT-B	**77.40**	**71.51**

TABLE X
COMPARISON OF CAM AND AUX CAM MIOU AT VARIOUS MASK THRESHOLDS

Mask_thre	CAM mIoU	AUX CAM mIoU
0.3	64.15	67.95
0.4	68.35	67.03
0.5	74.63	68.04
0.6	77.34	64.08
0.7	77.75	69.34
0.8	**78.62**	**71.51**
0.9	73.03	65.64

the board when using ViT-B as the self-supervised encoder backbone. However, the results obtained with ViT-S as the backbone do not surpass the baseline. This could be attributed to the unsuitability of finer-grained features from smaller encoders for this specific task context. Among them, DINO improved by 4.1%, while MAE and iBOT improved by 1.94% and 1.06%, respectively. A potential explanation is that we established a higher threshold for positive sample selection to achieve more precise patch relationships. However, MAE and iBOT employed the method of masked language modeling, which enabled the network to focus more on global information, thereby resulting in a higher overall similarity. In contrast, DINO focuses more on the semantic consistency of patch features, thereby feature similarities obtained from DINO are more suitable for the optimization task.

4) Hyperparameters: The hyperparameters in the experiment will have a great impact on the experimental results. In this part, we select the optimal hyperparameters through comparative experiments.

a) CAM ignore threshold: In the context of utilizing intermediate layer CAM to provide feature similarity information, the selection of a threshold is crucial for determining the criteria for sample pairwise relations, directly impacting the PTC loss within the CAM. To this end, we conducted a series of comparative experiments aimed at identifying the optimal threshold. As shown in Table X, the experimental results indicated that a lower threshold could lead to the propagation of incorrect similarity relations to the output layer, thereby reducing the accuracy of the output CAM, sometimes even falling below the performance of intermediate layer CAM. As the threshold increased, the confidence in the corresponding patch relations gradually improved, which in turn enhanced the precision of the CAM. However, when the threshold was set to 0.9, there was a noticeable decline in CAM accuracy. This could be attributed to the excessively high threshold reducing the number of patches involved in the calculation, thereby limiting the range of patches considered by the PTC loss and resulting in a decrease in accuracy.

TABLE XI
SEGMENTATION RESULTS AT DIFFERENT START VALUES

Start Value	mIoU (%)
0.2	80.01
0.5	**80.51**
0.8	79.79
1.0	62.60

b) Gradient clipping start value: Initiating gradient clipping too early can disrupt the model's ability to learn from clear data during the early stages of training, while clipping too late may result in the model being adversely affected by noise due to the incorporation of incorrect labels into the training process. Therefore, determining the appropriate timing for gradient clipping is particularly crucial. For ease of comparison, we opted not to use iterative training and post-processing operations in this part of the comparative experiments. Through experimental comparisons on the average gradient threshold for initiating gradient clipping shown in Table XI, we ultimately selected 0.5 as the criterion for initiating gradient clipping, thereby achieving optimal results.

V. CONCLUSION

This article proposes a comprehensive WSSS approach for remote sensing data. To tackle the challenges posed by the abundance of scene targets and minor distinctions between categories inherent in remote sensing data, we propose a method that leverages a self-supervised encoder fine-tuned on remote sensing datasets to provide additional feature similarity information, thereby effectively improving the quality of generated pseudo-labels and reducing the noise in the pseudo-labels. Considering that noise present in pseudo-labels can disrupt the training process of the segmentation network, we further propose a novel loss computation method that combines label smoothing and gradient clipping, effectively mitigating the adverse effects of noise on network training. Experiments conducted on the ISPRS Potsdam and Vaihingen datasets, demonstrate that our proposed method achieves SOTA results, with performance closely matching that of fully supervised methods. The results on the iSAID dataset indicate that our method can achieve SOTA results across various types of remote sensing data. Additionally, experiments on the natural dataset VOC2012 indicate that our proposed method can also achieve SOTA results, fully demonstrating the generalizability of our approach. In future work, we aim to explore the integration of additional self-supervised tasks to further improve feature representation and segmentation accuracy. Furthermore, the adaptation of our approach for multitemporal and multispectral remote sensing data will be studied, aiming to provide more comprehensive solutions for various applications.

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